



Aim

Comprehend the scope of different wine types arising from various grape varieties and learn how they are classified.

TYPES OF WINES

There are five main types of wines:

- Red still wine (table wines and traditionally drunk with a meal)
- White still wine (also known as table wine)
- Rosé wine (wine which is midway between white and red wine)
- Sparkling wine (can be served at any time)
- Fortified wine (traditionally drunk after dinner)

Red Wines

These include:

- **Light red wines** - these generally have quite subtle flavours and may be slightly perfumed. They have a lower alcohol content and are relatively easy to drink even for people who are not ordinarily wine drinkers, e.g., Pinot Noir, Gamay, and Cinsault.
- **Medium-bodied reds** - these have a mild to moderate flavour and tend to work better with food, e.g., Grenache, Merlot, Barbera, Sangiovese, and Cabernet Franc.
- **Full-bodied reds** - these have a rich dry flavour, higher alcohol content and more tannins, e.g., Cabernet Sauvignon, Shiraz and Malbec

White Wines

These include:

- **Sweet white wines** - these are often highly perfumed as well as sweet and their sweetness helps to balance out the bitterness of the grapes used, e.g., Riesling, Moscato, and Chenin Blanc.
- **Dry white wines** - these are generally light wines with a refreshing, dry, slightly tart or zesty taste, e.g., Chardonnay, Sauvignon Blanc, Verdejo, Soave, and Pinot Gris.
- **Full-bodied whites** - these have a fuller flavour similar to reds and undergo malolactic fermentation. Many are aged in oak and are usually better when about 3 to 4 years old though some can be stored for longer, e.g., Semillon, Viognier, and Oaked Chardonnay. Some blends of dry and full-bodied whites tend to taste more like full-bodied wines, e.g., Semillon Chardonnay and Semillon Sauvignon Blanc.



Rosé Wines

Although rosés are part red and part white, they are more similar to white wines and are usually served chilled. They are popular in the Mediterranean region.

- **Rosés** - these often have quite a dry taste and some popular ones include Pinot Noir Rosé, Sangiovese Rosé, Cotes Du Rhone Rosé, and Provence Rosé.

Sparkling Wines

Sparkling wines have evolved a lot over recent decades. No longer is the sector dominated by Champagne in the way that it once was. Sparkling reds are also quite popular these days and like their white counterparts they are served chilled.

- **Champagne** - this is made by adding a special type of yeast and rock sugar to still wine which introduces the bubbles as it ferments. Champagne can only be made in the Champagne region of France.
- **Other sparkling wines** - there are various other sparkling wines, some are made in the champagne style, some are made by forced carbonation techniques, others are made by fermenting wines under pressure, e.g., Lambrusco and Prosecco.

Fortified Wines

These are wines which have had either brandy or grape spirit added to them and include dessert wines which tend to be sweeter although there is no particular distinction between fortified and dessert wines. The addition of pure spirit increases the alcohol content from 11 or 12% to about 18 to 20%.

- **Fortified wines** - some are more usually drunk after a meal, e.g., Port, Madeira, Sherry, Marsala, and Vermouth. Others are more usually drunk before a meal, e.g., Dry Sherry.
- **Dessert wines** - these tend to be sweeter, e.g., Tokaji (Tokaji), Sweet Sherry, Sauternes, and Spätlese.

SELECTING WINE GRAPES

When selecting what to grow, the prime concern should be the varieties' suitability to your locality. Large vineyards can experiment but small operations must carefully consider their selection.

Many wineries today do not grow all of their own grapes. They often buy grapes from growers in their district, or surrounding districts. If you locate a new vineyard in a region that already has large well-established wineries doing this, you are enhancing the prospect for sales. Even if you do have a market for your produce, there may be occasional boom years, or years when market conditions change, and to have the possibility of selling to other wineries, not too far away, can be a big advantage. Similarly, if you are not able to produce sufficient grapes for your own needs one year, you may still have a possibility of buying grapes from other vineyards in your locality.

Consider:

Chardonnay: ripens early and is not normally damaged by rain.

Chenin Blanc: ripens early to mid-season; is susceptible to splitting by rain and to bunch rot and should be grown only in warm areas.

You should make comparisons like this between all the varieties you are considering.

Yield

The yield per acre should be considered when choosing what varieties to plant. This will affect the economic viability of your operation. The following varieties will likely perform as outlined below in areas with low to medium rainfall and good soil.



Variety	Tonnes Per Acre in 475 mm rainfall	Tonnes Per Acre in 575 mm rainfall
Cabernet Sauvignon	2 - 5	6 - 11
Shiraz	3 - 6	7 - 14
Grenache	4 - 10	8 - 16
Rhine Riesling	2 - 5	6 - 11

2 tonnes per acre would be considered low yield.

10 tonnes per acre would be considered high yield.

Low yield varieties include Cabernet Sauvignon, Gewürztraminer and Riesling.

Low to medium yield varieties include Chardonnay, Pinot Noir, Rieslaner and Sauvignon Blanc.

Medium yield varieties include Ehrenfeleser, Malbec, Merlot, Meunier, Muscat, Perle, Pinot-gris, Reichensteiner and Semillon.

Medium to high yield varieties include Chasselas, Chenin Blanc, Faber, Kerner, Malbec, Pinot Blanc, some new Riesling cultivars, Siegerebe and Sylvaner.

High yield varieties include Bacchus, Kerner, Muller-Thurgau, Mario Muscat and Zweigeltrebe.

Note the indication of yields above is a generalisation. The locality they are grown in can also strongly affect the yield. Mueller-Thurgau, for example, is a low yield variety in Britain, but high yield in some other localities.

WINE GRAPE VARIETIES

Chenin Blanc

A premium white wine in France. A quality wine in the dry, sweet or sparkling style. Late harvest grapes.

Characteristics: Produces a light wine that retains a pronounced varietal character with medium acidity. Late harvest grapes give a distinctive wine with complex character.

Culture

Soils - Calcareous soils are said to improve quality considerably in France, although this has not been confirmed in South Africa or California where it appears less demanding in relation to the soil.

Vigour - Vigorous.

Pruning - Medium to long canes with a high proportion of old wood to new canes (cordons).

Disease susceptibility - Downy mildew - medium to high, Powdery mildew - moderate to high, Bunch rot and *Botrytis* - moderate to high.

Wet weather – Susceptible.

Rootstocks - Use those with moderate vigour.

Yields - Moderate to high.

Chardonnay (Pinot Chardonnay)

A premium quality white wine variety of central Europe. Has a distinctive wide open, basal sinus and yellow-green grapes. In soils lacking calcium the wine is less aromatic.

Characteristics: The wine has a fine and distinctive varietal aroma and flavour.

Culture

Soils - Likes well drained, fertile but drier calcareous soils. Not suited to heavy clay loams. Heavy nitrogen fertilisers not recommended since set becomes poor and grapes are more susceptible to rots.

Vigour – Moderate.

Pruning - Medium to long cane pruning.

Disease susceptibility - Downy mildew - low, Powdery mildew - fair, Bunch rot and *Botrytis* - moderate to high.

Wet weather - Susceptible to rain at harvest.

Rootstocks - Vigorous ones preferred.

Yields - Low to moderate - depending on clone and climate.

Pinot Blanc

A premium white French variety. Early ripening grapes.

Characteristics: The wine retains fair acidity with fine varietal flavour and aroma.

Culture

Soils - Likes well drained stony or calcareous soils. It has resistance to drought.

Vigour – Moderate.

Pruning - Use long canes.

Disease susceptibility - Downy mildew - low, Powdery mildew - fair to high, Bunch rot and *Botrytis* - fair to high.

Wet weather - Fair resistance.

Root stocks - Does well on vigorous root stocks.

Yields - Moderate to high.

Semillon

A white grape grown throughout France for its fine and dry or late harvested sweet wines of considerable finesse and aroma. In Australia it is known as Hunter Riesling and White Madeira; in South Africa it is known as Green Grape.

Characteristics: Soft, fair acidity in cool climates, and a distinctive herbaceous aroma and flavour.

Culture

Soils - Prefers lighter soils.

Vigour - Moderate to vigorous.

Pruning - Use long canes and summer prune in more humid areas.

Disease susceptibility - Downy mildew - fair, Powdery mildew - fair to high, Bunch rot and *Botrytis* - high, but see comments under Sauvignon Blanc.

Wet weather - Poor resistance, also susceptible to sunburn.

Yields - Moderate.

Muscat Ottonel

A quality white grape grown throughout Europe. Early ripening. Short internodes are typical.

Characteristics: Low acidity, fine aroma, pronounced varietal flavour.

Culture

Soils - Deep fertile soils are best; calcareous soils induce chlorosis. Drought conditions produce insufficient growth.



Vigour - Low to moderate with a tendency to spread rather than grow upright, therefore a smaller system such as head cane are recommended.

Pruning - Use short canes.

Disease susceptibility - All diseases - fair.

Wet weather - Moderately susceptible.

Root stocks - Moderate to vigorous. Choose those like 26-G or SO-4 which improve set: an inherent problem with this variety.

Yields - Moderate, but consistent in cropping.

Muscadelle

A widely planted white grape variety throughout the world. Young spring growth is reddish. Fruit ripens early.

Characteristic: Marked bouquet somewhat reminiscent of Muscat.

Culture

Soils - Well drained, deep profiled soil.

Vigour - Fairly vigorous.

Pruning - Use medium canes.

Disease susceptibility - All diseases - moderate.

Wet weather - Does not appreciate hot humid weather.

Yields - Moderate.

Gewurztraminer (Savagnin Rosé)

Produce both white and pink berries in the one bunch. Matures early. Leaf is distinctive, being small, round, dark green and rough with shallow lobes. This is a clonal selection of the Traminer variety.

Characteristics: Distinctive spicy wine, pronounced varietal aroma and flavour, and low acidity.

Culture

Soils - Likes deep fertile soils, use nitrogen with caution since excess causes poor set. Has a tendency to exhibit chlorosis in calcareous soils.

Vigour – Moderate, has short internodes, lacks upright growth. There is a tendency for multiple shoots to be produced from the buds.

Pruning - Use long canes with a higher proportion of older wood in a cordon system. Summer thinning of the shoots is advisable.

Disease susceptibility - Downy mildew - fair, Powdery mildew - high, Bunch rot and *Botrytis* – high.

Wet weather - Susceptible to wet weather at flowering and ripening, causing poor set and splitting of berries.

Root stocks - Low to moderate vigour preferred.

Yields - Low to moderate.

Cabernet Sauvignon

A premium black grape. In cool climates it ripens late in the season, giving small to medium yields of high-quality fruit. This grape will not ripen fruit in very cool districts. The quality of the wine compensates for the generally low yields. Leaf is distinctively dark green and rough on the surface with deep sinuses; reddish shoot growth in spring is typical.

Characteristics: Fullness of varietal, depth in bouquet and flavour, coupled with high acidity and tannin content.



Culture

Soils - Suits most soils.

Vigour – Vigorous.

Pruning - Medium to long cane pruning preferred with a high proportion of old wood to young canes (use cordon).

Disease susceptibility - Downy mildew - fair, Powdery mildew - high, Bunch rot and *Botrytis* - low.

Wet weather - Good resistance to wet weather at harvest.

Root stocks - Low to moderate vigour are best to control vigour and improve set.

Yield - Low but can be increased with good pruning.

Carignan

One of the most widely grown red wine grapes in the world. Has large lobed leaves with hairs on the underside. Fruit matures late.

Characteristics: Moderate colour, good tannin and no pronounced varietal character.

Culture

Soils - Sandy soils to most soil types.

Vigour - Vigorous and upright.

Pruning - May need summer pruning.

Disease susceptibility - All diseases - low.

Wet weather - Fruit not readily damaged by rain during ripening.

Yields - Up to 20 to 30 tonnes per hectare is possible.

Vitis rotundifolia

This is an American species, commonly known as Muscadines, Scuppernongs or "Scupps". Scupps differ to *Vitis vinifera* in a number of ways:

- Highly resistant to disease
- Larger, looser bunches
- Berries can be larger (to 50mm diameter)
- Berries in a bunch ripen progressively (not all at once)
- Berry colour is much more variable than *V. vinifera*
- Berry skin is loose
- Taste is different to *V. vinifera* (some people say a little like pineapple flavour)
- Plants are either male or female

Cultivated named varieties of Scupps are available in nurseries throughout North America, but not everywhere else in the world. In many countries (e.g., Australia) it is difficult to find cultivars of this species, and even importing them can be a problem due to quarantine regulations.

Importing seed may be easier than importing plants, but the reliability of seedlings to provide desirable fruit is uncertain. You may need to grow and select the best plants, if attempting to establish *V. rotundifolia* plantings this way.

WINE GRAPES

For dry or table wines, high acidity and moderate sugar content are desired. Grapes with high sugar content and moderately low acid are required sweet or dessert wines. Grapes such as Cabernet Sauvignon, Riesling and Pinot Noir all have outstanding bouquet and flavour essential for production of the highest quality premium wines. Thin-skinned varieties will require more care during harvest and post-harvest techniques than those with thick skin and a firmer pulp.

OAKING

Wines can also be differentiated based on their degree of oaking:

- Oaked wine
- Lightly oaked wine
- Unoaked wine

Oaked Wines

Oaked wines, also known as wooded wines, are aged in oak barrels prior to bottling. Oak barrel alternatives, such as oak chips, can also be used. The oak enhances the colour of the wine, softens its acidity, and adds aroma compounds (oak lactones and vanillin), giving the wine a richer taste. Different types of oak can be used, imparting distinct characteristics, e.g., French oak, American oak and Hungarian/Eastern European oak. The release of aromas can be increased by toasting the oak with fire.

The length of time a wine sits in oak determines its degree of oaking. The longer a wine sits in oak, the more oak aroma compounds are released into the wine and the stronger these aromas become. If a wine is aged in oak too long, the oak aromas tend to overpower the fruit aromas of the wine. This wine is said to be unbalanced.

The flavour profile imparted by the oak depends on the grapes it is paired with.

Most red wines and several white wines, such as Chardonnay, are oaked. Oaking generally darkens the colour of the wine. This is particularly visible in white wines which become a darker yellow the longer they sit in oak.

Oaked wines are often a little thicker than unoaked wines, as the oak barrels they are stored in are usually not completely airtight and slowly let water evaporate from the wine (approximately 10 % per year).

Mainly due to the additional costs associated with production, oaked wines are typically more expensive than unoaked wines. Production costs can be reduced by adding oak essence to the wine; however, wine produced this way cannot compete in complexity with wine properly aged in oak.

Unoaked Wines

Unoaked wines, also known as unwooded or naked wines, are matured without oak, and therefore do not contain oak aroma compounds. Their taste is predominantly determined by the fruit aromas of the wine. These wines are lighter, contain more acidity and taste fresher.

White wines, such as Riesling and Sauvignon Blanc, are mostly unoaked. These wines are best consumed while young.

